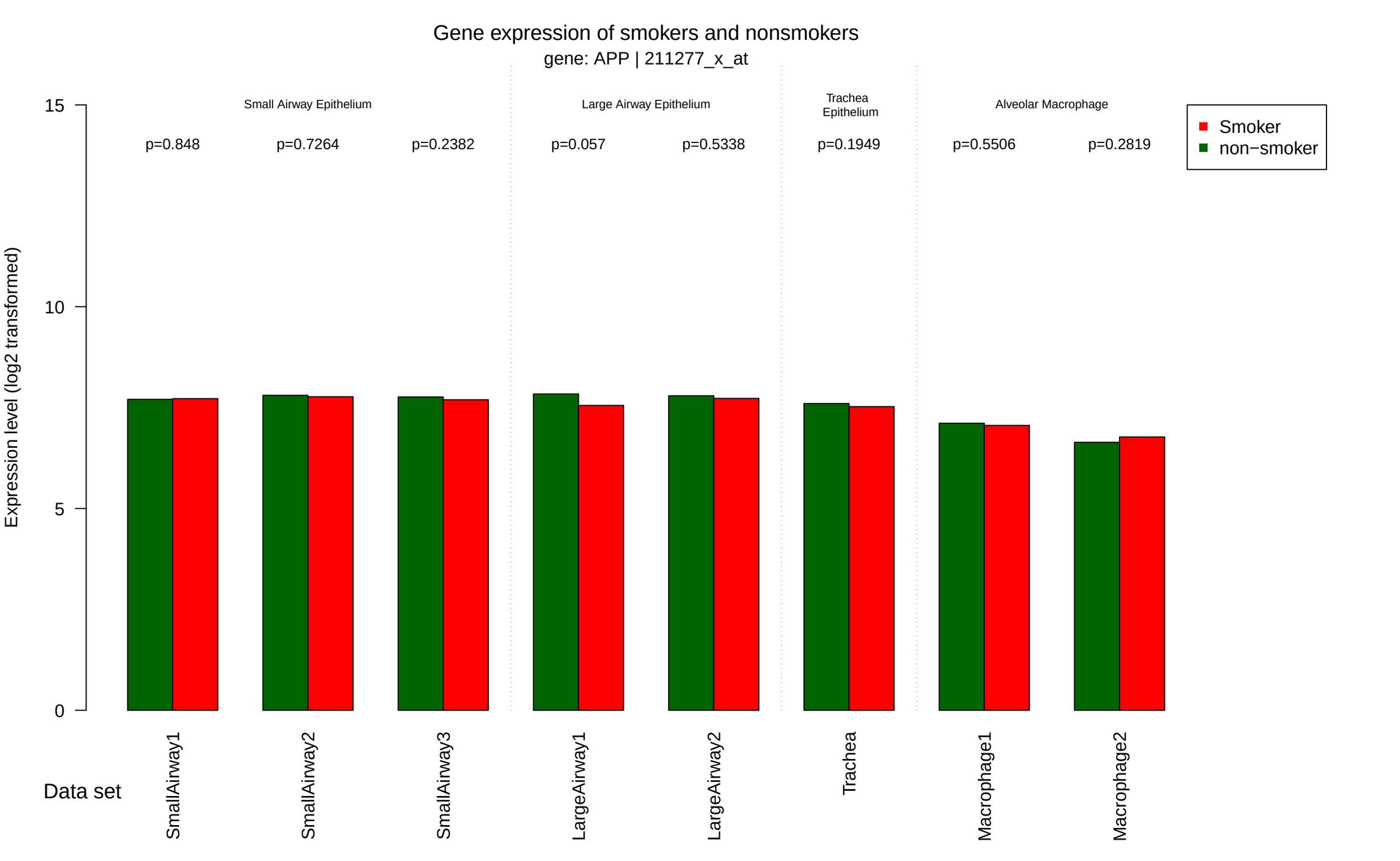
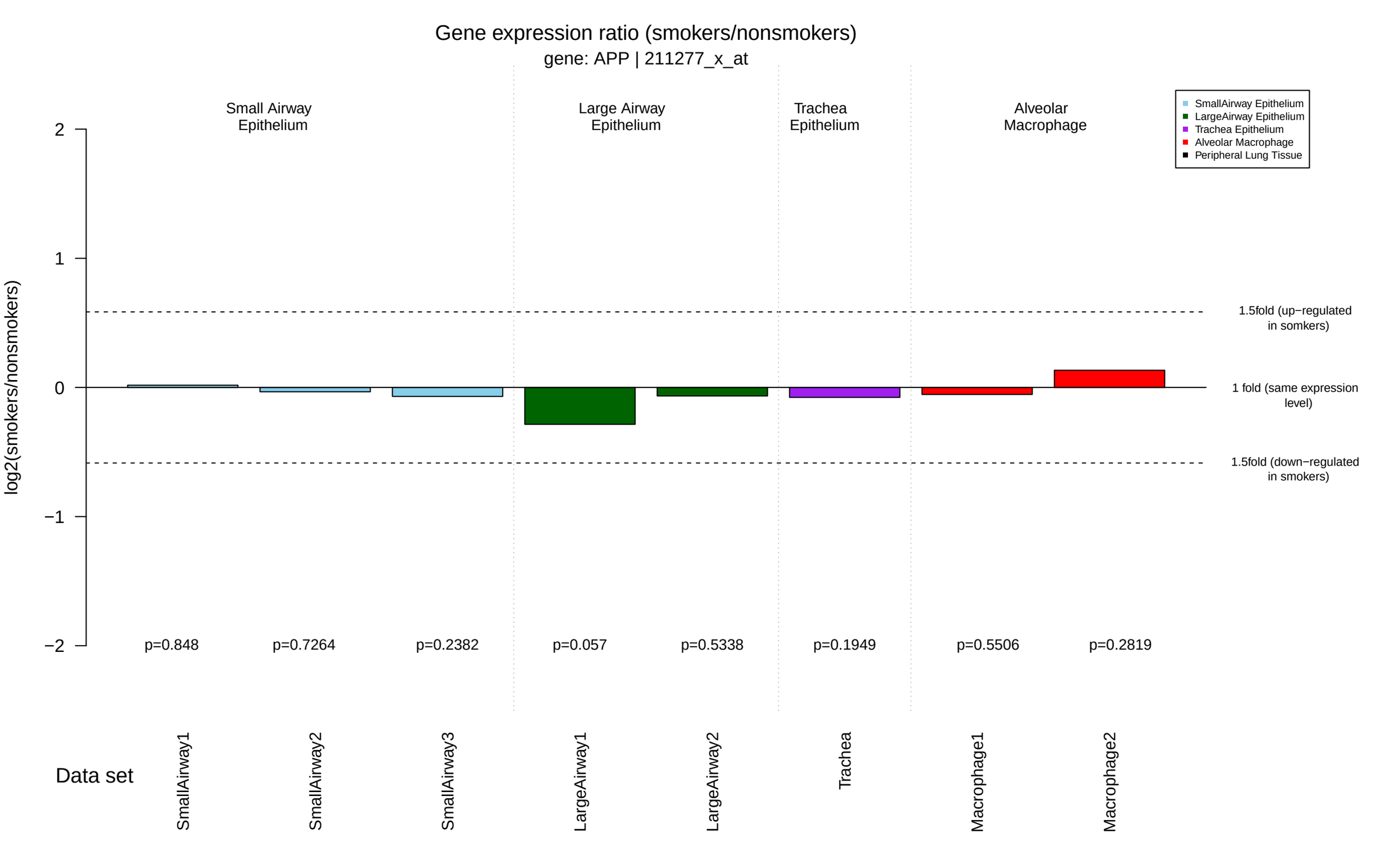
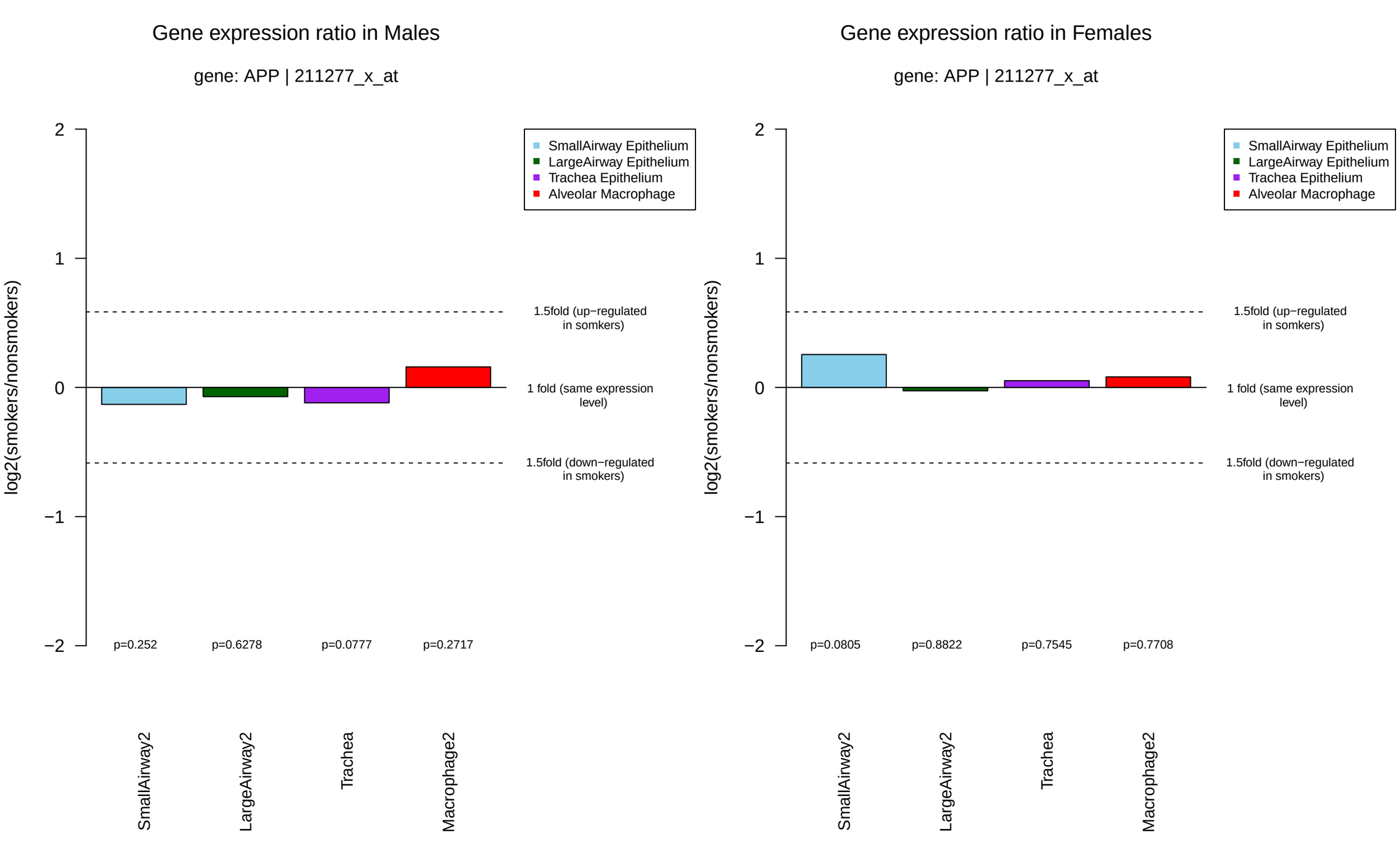
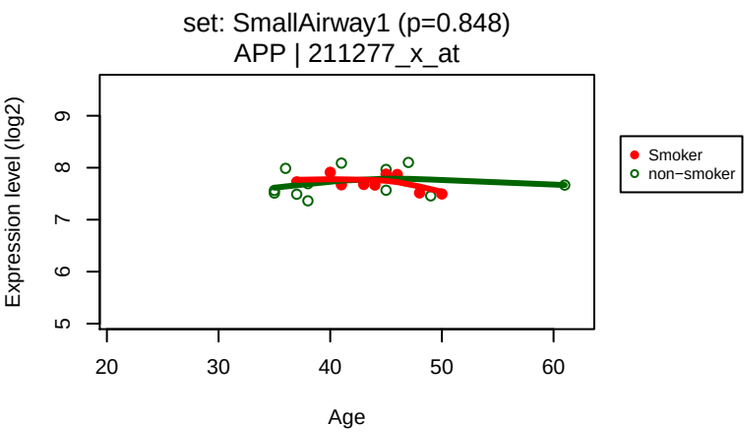
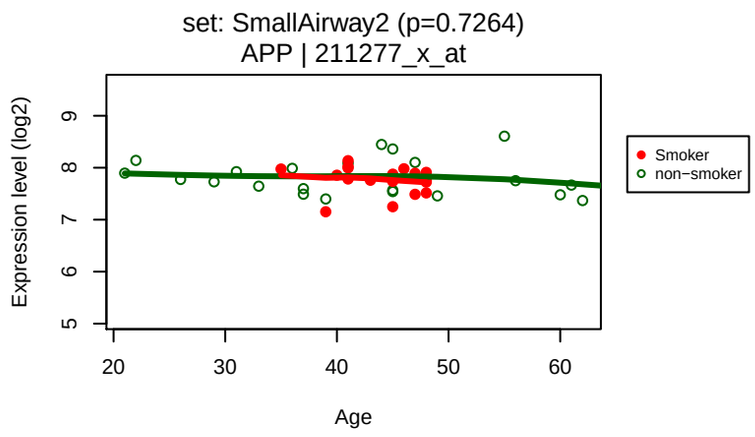
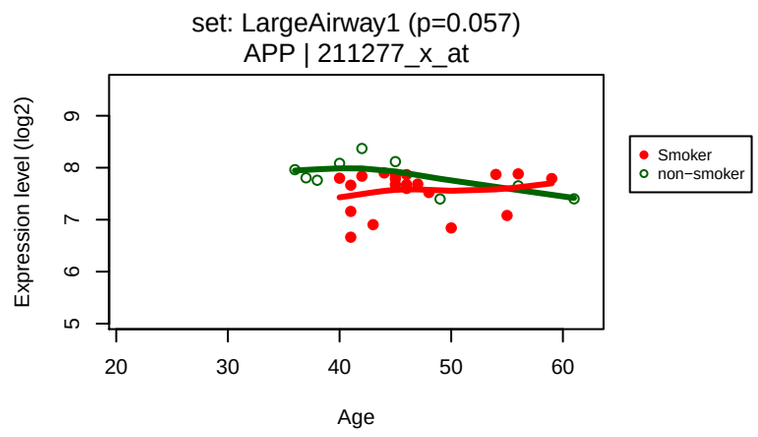

Gene expression of smokers and nonsmokers
gene: APP | 211277_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.848
p=0.7264
p=0.2382
p=0.057
p=0.5338
p=0.1949
p=0.5506
p=0.2819
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

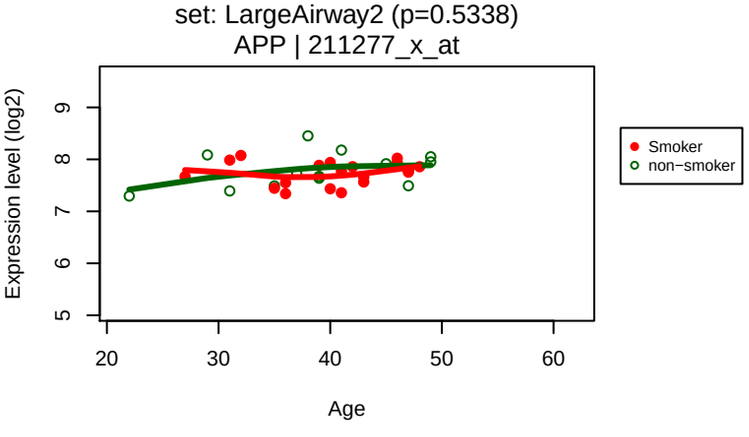

Gene expression ratio (smokers/nonsmokers)
gene: APP | 211277_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.848
p=0.7264
p=0.2382
p=0.057
p=0.5338
p=0.1949
p=0.5506
p=0.2819
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

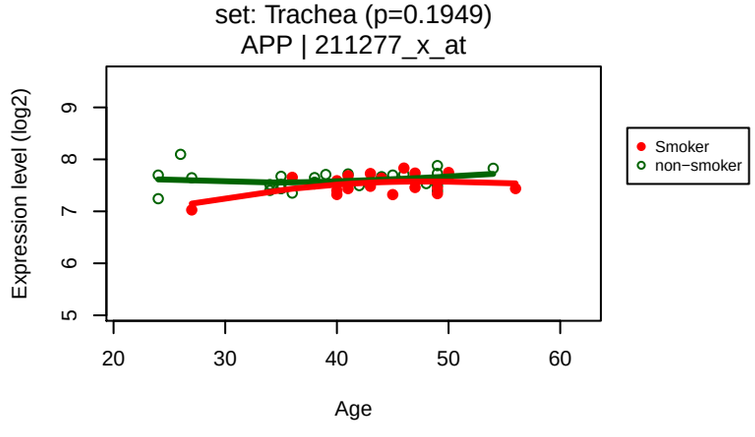

Gene expression ratio in Males
gene: APP | 211277_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.252
p=0.6278
p=0.0777
p=0.2717
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APP | 211277_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0805
p=0.8822
p=0.7545
p=0.7708
SmallAirway2
LargeAirway2
Trachea
Macrophage2

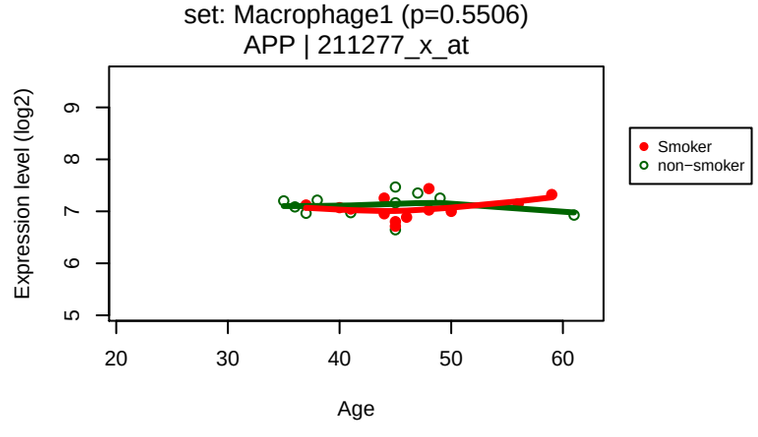

set: SmallAirway1 (p=0.848)
APP | 211277_x_at
Expression level (log2)
Age
Smoker
non−smoker

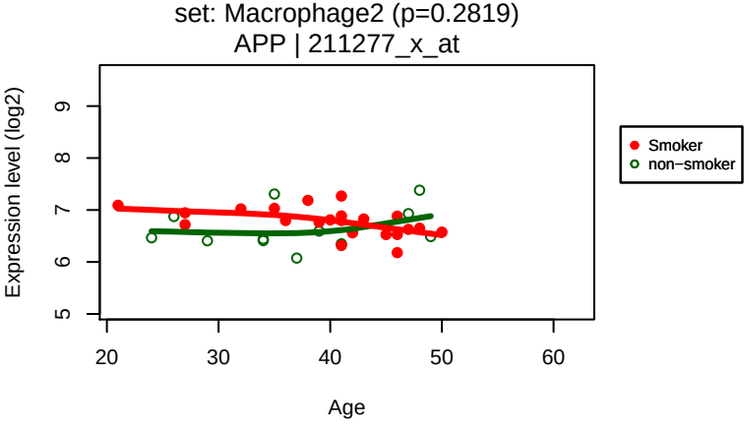

set: SmallAirway2 (p=0.7264)
APP | 211277_x_at
Expression level (log2)
Age
Smoker
non−smoker

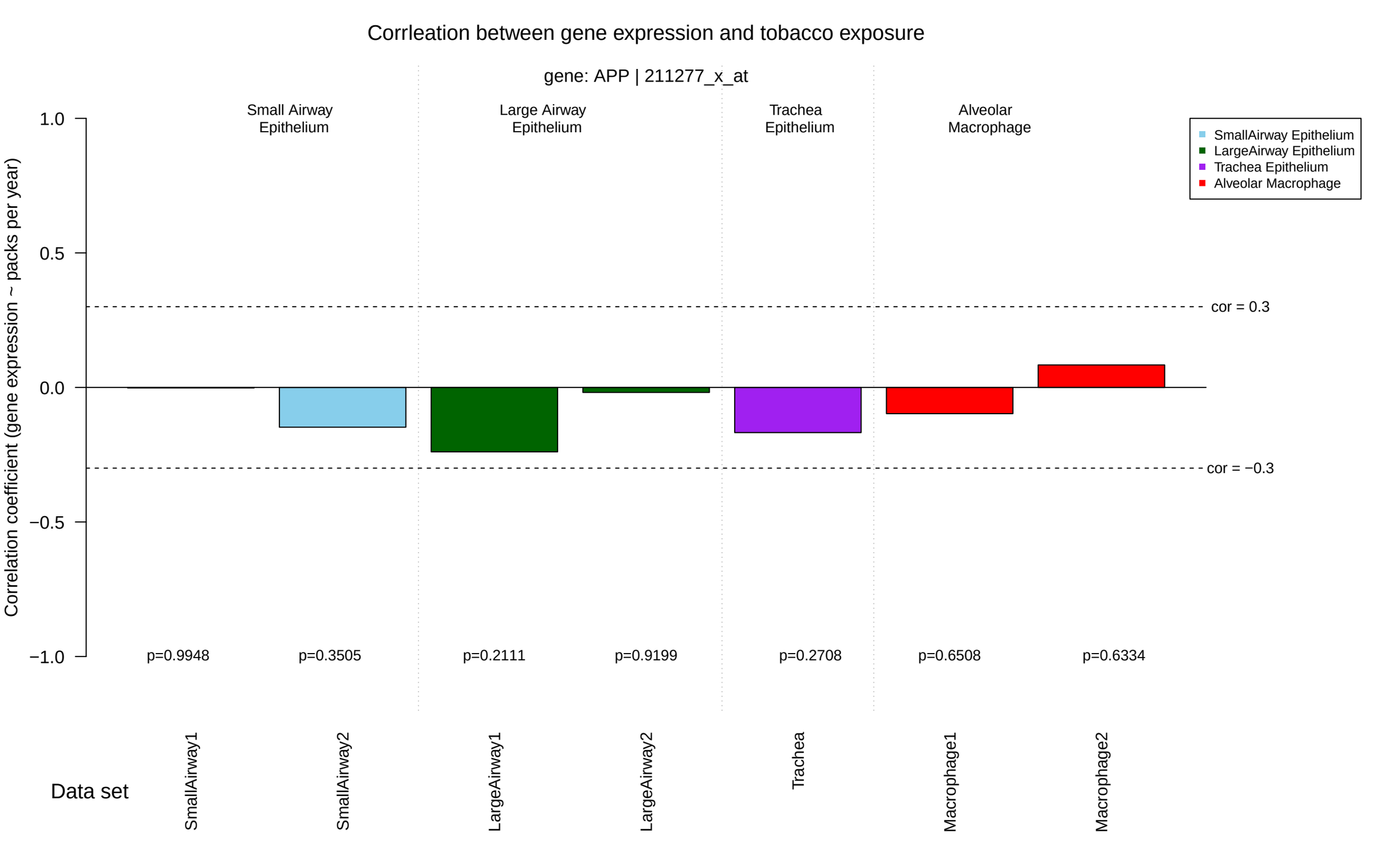

set: LargeAirway1 (p=0.057)
APP | 211277_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5338)
APP | 211277_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1949)
APP | 211277_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5506)
APP | 211277_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2819)
APP | 211277_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APP | 211277_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9948
p=0.3505
p=0.2111
p=0.9199
p=0.2708
p=0.6508
p=0.6334
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2